

Spam SMS Detection

Professionalized machine learning repository report

Problem

Classify SMS messages as spam or ham with a cleaned NLP pipeline and TF-IDF based models.

Workflow

Raw dataset -> preprocessing -> feature engineering -> model comparison -> evaluation outputs -> saved model -> Streamlit app.

Models Compared

Model	Purpose
Naive Bayes	Baseline comparison candidate
Logistic Regression	Baseline comparison candidate
Linear SVM	Baseline comparison candidate

Dataset Columns

Column
v1
v2
clean_message
target

Repository Improvements

- Organized folders for data, notebooks, source code, models, outputs, app, docs, images, and reports.
- Moved reusable logic into modular Python files.
- Added config.yaml for reproducible parameters.
- Added logging-oriented training scripts and output artifacts.
- Added documentation for business context, architecture, data dictionary, deployment, and usage.

Key Notes

- Spam messages often contain urgency, rewards, calls to action, and promotional language.
- The app highlights suspicious keywords and reports confidence.

Next Steps

- Run training after installing dependencies.
- Capture Streamlit screenshots for the images folder.
- Add SHAP or feature contribution explanations if presentation time allows.